

Diagnosing Where a Retailer Loses Its Customers

End-to-end analysis of 99,441 orders separating the fixable problem (delivery) from the structural one (retention). Built on the Olist Brazilian e-commerce dataset; "Atlas & Vine" is a narrative framing. Figures in reais (R\$).

The problem

A mid-sized online fashion retailer was seeing rising returns, falling repeat purchases, and a growing support backlog, with no clear read on where customers drop off or why. The brief: find the drop-offs, root-cause the frustration, and recommend what to actually do.

Approach

Cleaned and joined the transactional data, translated and labeled reviews, trained sentiment and theme classifiers, reconstructed a fulfillment funnel, segmented customers, and built a repeat-purchase model, then surfaced all of it in an interactive Streamlit dashboard and a business report.

What the data showed

Late delivery is the clearest lever.

Average review falls from 4.21 (on time) to 2.27 (late), a **1.94-star** drop. **62%** of late orders score 1-2 stars. Full-population, not a sample.

Retention is structurally low.

Only **3.1%** of customers ever order again. This is the headline business problem, bigger than delivery.

Experience does not predict repeat.

A leakage-free model (XGBoost, first-order features) scored ROC-AUC 0.59. The low score is the finding: it proves repeat behavior can't be targeted per customer from experience signals.

~3% of orders never arrive.

2,965 orders drop out of fulfillment, the biggest leak at carrier handoff. Customers who paid and got nothing.

The two findings that matter point opposite ways: fix delivery for everyone, and stop trying to predict retention for anyone.

Recommendation

Fix delivery as a system (target the carrier-handoff leak and the 22-day tail where satisfaction collapses), protect the high-value segment that drives ~57% of revenue, and do **not** fund a per-customer churn program the data can't support. The report sizes the addressable population rather than publishing a fabricated impact percentage, and points to a controlled delivery pilot as the honest next step.

BY THE NUMBERS

Orders analyzed	99,441
Unique customers	96,096
Reviews scored	11,997
Late-delivery star drop	1.94
Repeat rate	3.1%
Revenue in top segment	~57%

STACK

- Python, pandas, scikit-learn, XGBoost
- TF-IDF + LogReg classifiers (84% sentiment accuracy)
- TF-IDF + K-means topic clustering
- K-means segmentation + descriptive CLV
- Streamlit + Plotly dashboard

HONEST BY DESIGN

- Methods labeled accurately (K-means topics, keyword emotion, descriptive CLV)
- Oversampled complaint corpus never used for population rates
- Dashboard is interactive, not live
- No fabricated business-impact figure